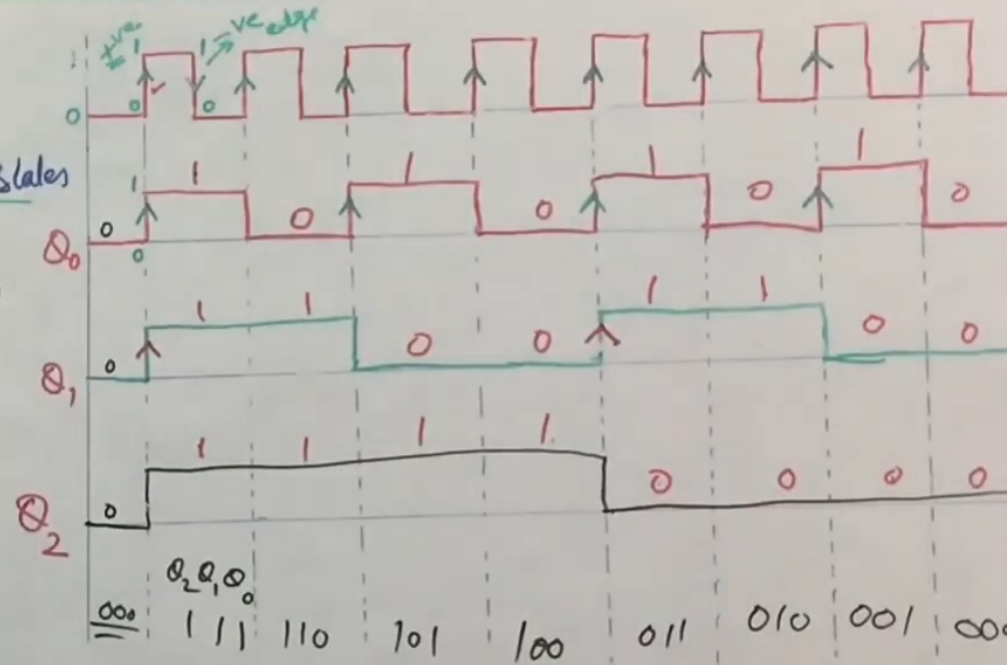
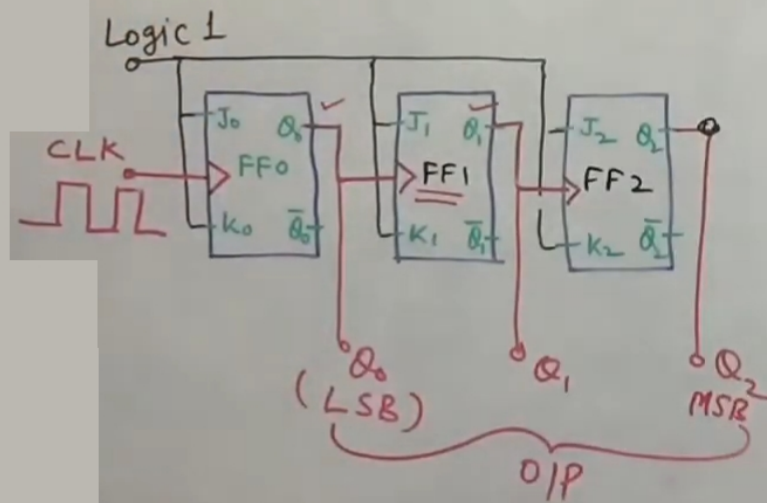


3-Bit Asynchronous Down Counter

3 bit $\rightarrow$ 3 Flip Flop

No. of states: $N = 2^n$
 $n = 3, N = 2^3 = 8 \text{ states}$

✓ MOD-8 Counter
(Max Count) = $8-1 = 7$

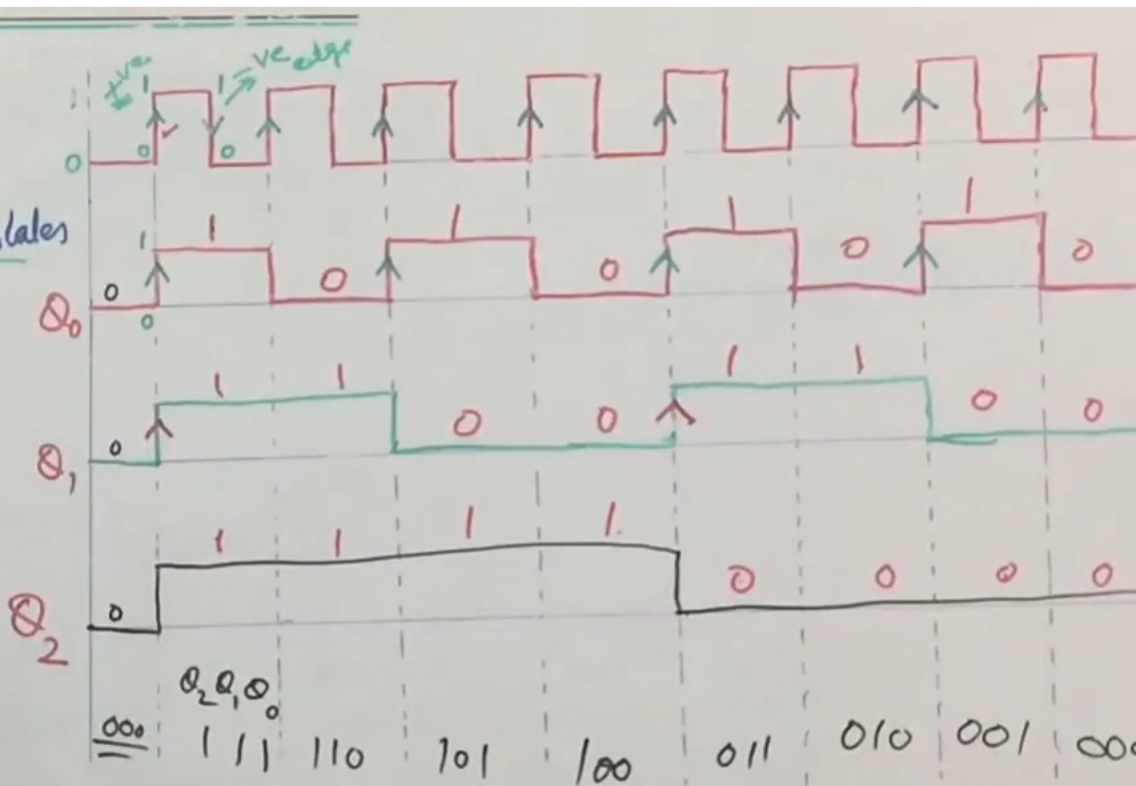


$$f_z = \frac{f_c}{8}$$

7 $\rightarrow$ 6 $\rightarrow$ 5 $\rightarrow$ 4 $\rightarrow$ 3 $\rightarrow$ 2 $\rightarrow$ 1 $\rightarrow$ 0

Descending order
Down Counter

Wave Form :-



⑦ → ⑥ → ⑤ → ④ → ③ → ② → ① → ⑦

Descending order

Down Counter

State Table :-

clk pulse	Q_2	Q_1	Q_0	Dec	CL
Initial	1	1	1	7	
1 st ($\uparrow$)	1	1	0	6	
2 nd ($\uparrow$)	1	0	1	5	
3 rd ($\uparrow$)	1	0	0	4	
4 th ($\uparrow$)	0	1	1	3	
5 th ($\uparrow$)	0	1	0	2	
6 th ($\uparrow$)	0	0	1	1	
7 th ($\uparrow$)	0	0	0	0	

Trick to consider UP or Down Counter :-

